

ShowUp Platform Documentation

Next-Generation Collegiate Event Discovery, Ticketing, UPI Payments, and Fest Management Ecosystem

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INSTITUTIONVellore Institute of Technology
(VIT), Vellore**LIVE PRODUCTION URL**showup.vercel.app ↗**GITHUB REPOSITORY**[github.com/raynaagarwal01-
rgb/showup](https://github.com/raynaagarwal01-rgb/showup) ↗**GRAVITAS '26 HUB**[/fests/gravitas26](https://fests/gravitas26) ↗

1. Executive Summary & Project Mission

At large university campuses such as **VIT Vellore**, thousands of students engage in hackathons, technical masterclasses, esports tournaments, and cultural competitions across major festivals like **graVITas** and **Riviera**. However, the existing registration and discovery process is fragmented across scattered WhatsApp links, Google Forms, and manual check-in sheets.

The Solution — ShowUp: An end-to-end modern web application providing instant event discovery, team formation with secret join codes, online UPI and card payment processing, automated high-contrast offline passes (PNG photo pass), client-side 1920x1080 participation certificate generation, real-time organizer announcement broadcasts, per-event Q&A discussion forums, calendar synchronization (Google & iCal), and comprehensive On-Duty (OD) attendance export for university administration.

2. Complete Page-by-Page & Button-by-Button Guide

Page 1: Landing Page (/)

The gateway welcoming students and organizers to the campus event ecosystem.

"Explore Events" Button

Directs users to the live directory (/events) with full search and filters.

"Host an Event" Button

Navigates club leads to the Organizer Studio (/organizer) to publish events.

"Sign In / Get Started" Buttons

Launches student & organizer authentication flows with immediate session persistence.

Category Pill Tiles

Instant shortcuts filtering Hackathons, Workshops, Cultural, and Competitions.

Page 2: Events Directory Page (/events)

The centralized search and discovery engine featuring multi-dimensional filtering.

Live Keyword Search Input

Real-time instant filtering matching event title, tagline, club name, venue, or college.

Scope Segmented Control ("All" / "Internal" / "External")

Filters events restricted to campus students versus open to outstation colleges.

Category Badges Row

One-click pills: "All", "Hackathon", "Workshop", "Competition", "Cultural", "Talk".

"Saved (count)" Wishlist Pill

Filters the view to only bookmarked events shortlisted by the student.

From-Date & To-Date Calendar Range Pickers

Narrows down events scheduled between specific dates.

Sort Dropdown Menu

Sorts by "Soonest First", "Latest First", "Fee: Free First", "Fee: Low to High", etc.

Event Card Bookmark Button (Heart/Ribbon)

Toggles immediate saving of the event to local storage without page reload.

Page 3: Event Details & Registration Page (/events/:id)

Comprehensive event portal with interactive participation tabs and registration actions.

Tab: "Overview & Rules"

Tab: "Announcements"

Displays complete formatted description, eligibility, and numbered contest rules.

Displays real-time organizer broadcast feed with glowing URGENT ALERT badges.

Tab: "Q&A Forum"

Student discussion thread with "Ask a Question" box and verified organizer replies.

"Register" / "Create Team" / "Join with Code"

Handles solo registration, team creation with unique join code, or joining via code.

"Download Pass (PNG)" Button

Generates an 800x1080 high-contrast photo pass with QR code for offline door entry.

"Download / Preview Certificate (PNG)" Button

Renders high-res 1920x1080 participation certificate with ShowUp seal & verification QR.

"Add to Calendar" Dropdown

Directly exports to Google Calendar or downloads .ics file for Apple & Outlook.

Page 4: Official graVITas '26 Dedicated Fest Hub (/fests/gravitas26)

Exclusive portal dedicated to VIT Vellore's flagship international technological festival.

Live Countdown Clock

Tick-by-tick countdown tracking Days, Hours, Minutes, and Seconds until Sep 26, 2026.

Schedule Day Switcher Tabs

"All 3 Days (56)", "Day 1 · Sep 26", "Day 2 · Sep 27", "Day 3 · Sep 28".

"Teams Only" & "Free Entry" Toggles

Instant toggles to filter team hackathons or free entry competitions.

OD & Accommodation Accordion

Verified information on on-duty attendance, hostel stay, and venue check-in.

Page 5: Student Dashboard / My Events (/dashboard)

Personalized student cockpit managing all registered event passes.

"Show / Hide Ticket" Button

Expands scannable QR ticket on screen for door check-in staff.

"Download Offline Pass (PNG)" Button

Saves the official pass image straight to mobile camera roll or computer.

"Certificate" Button

"Cancel Registration" (X Icon)

Opens the verified participation certificate modal for high-res PNG export.

Cancels ticket and automatically releases seat back to the waitlist / pool.

Page 6 & 7: Organizer Studio & Attendance OD Export (/organizer)

Dedicated back-office suite for university club organizers.

"Create Event" Wizard

Publishes new events with custom rules, banner hues, team limits, and fees.

"Export OD & Attendance (CSV)" Button

Generates a 13-column academic OD spreadsheet for VIT administration approval.

"Door Check-in Scanner" (Webcam QR)

Live camera scanner verifying participant tickets at venue entrance in real-time.

3. Tools & Technologies Breakdown: Who, Why & How

ShowUp is built following modern engineering best practices, leveraging industry-standard frameworks, libraries, and cloud infrastructure:

Technology / Tool	Who Created It	Why It Is Used	How It Is Used in ShowUp
React 19	Meta / Open Source Community	Component-driven, high-performance user interface with modern concurrent rendering.	Powers all interactive pages, modals, tab switchers, and real-time state management.
TypeScript 6	Microsoft	Static type safety; eliminates null pointer exceptions and guarantees data schema integrity.	All models (<code>EventRecord</code> , <code>Registration</code> , <code>Announcement</code> , <code>EventQuestion</code>) are strictly typed.
Vite 8	Evan You	Ultra-fast HMR dev server and optimized Rollup production bundler.	Compiles and bundles the application into optimized minified CSS/JS chunks.
Tailwind CSS v4	Tailwind Labs	Utility-first responsive CSS engine for elegant dark-mode aesthetics without bloat.	Implements custom theme palettes (<code>bg-ink</code> , <code>bg-surface</code> , <code>text-coral</code>) and mobile breakpoints.
Supabase (PostgreSQL 15)	Supabase Inc.	Cloud relational database with Row Level Security (RLS) and real-time persistence.	Stores 104+ events (including 56 graVITas events), user registrations, announcements, and Q&A.
HTML5 Canvas 2D API	W3C Web Standards	Zero-dependency client-side graphic rendering directly in the user's browser.	Synthesizes 1920x1080 participation certificates and 800x1080 offline pass PNGs offline.
node-qrcode	Soldair / Open Source	Fast cryptographic QR matrix generation.	Generates dynamic ticket validation QRs and UPI payment checkout QR codes.

Technology / Tool	Who Created It	Why It Is Used	How It Is Used in ShowUp
html5-qrcode	Minhaz / Open Source	Camera-based QR code decoding engine.	Provides the live webcam scanner in the Organizer Door Check-in module.
Lucide React	Lucide Project	Consistent, accessible vector iconography.	Renders all navigation, action, alert, and status icons across the entire UI.
Vercel Edge Cloud	Vercel	Global content delivery network (CDN) and automated Git deployment.	Hosts the live application with automatic SSL and zero-downtime rolling deploys.
GitHub	Microsoft	Git source code version control and CI/CD triggers.	Maintains complete codebase history at <code>raynaagarwal01-rgb/showup</code> .

4. Key Architectural & Functional Innovations

- **Dual-Mode Resilience (Cloud & Offline Demo Db):** If internet connection drops or API keys are missing, the system gracefully degrades to the local Indexed/LocalStorage demo database without crashing.
- **Instant Client-Side Certificate Synthesis:** Eliminates costly PDF rendering servers by generating 1080p certificates on HTML5 Canvas in under 80 milliseconds.
- **Official Academic OD Attendance CSV:** Standardized to VIT Vellore administrative guidelines for automated submission to Student Welfare Offices.
- **Comprehensive UPI Payment Gateway Simulation:** Realistic checkout modal supporting GPay, PhonePe, Paytm, and QR Scan with zero processing fees.

5. Project Verification Checklist

- ✓ Live Production URL: <https://showup.vercel.app>
- ✓ graVITas '26 Hub: <https://showup.vercel.app/fests/gravitas26>

- ✓ **GitHub Repo:** <https://github.com/raynaagarwal01-rgb/showup>
- ✓ **Build Status:** Compiled with 0 TypeScript/Vite errors (commit `b0e0486`)
- ✓ **Student Credential:** Rayna Agarwal (Reg: 25BCE0703), VIT Vellore

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